OMCS6250
Computer Networks

Intradomain Routing

With Instructor Dr. Maria Konte

Control Plane, Intradomain Routing,
Link-State vs Distance Vector Routing Algorithms

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Learning Objectives

- **Transport vs. IP**: where routing fits
- What routers do: **routing vs. forwarding**
- Distinguish **intradomain** vs. **interdomain** routing
- Compare **link-state** vs. **distance-vector**
- Walk through how each approach computes routes (Dijkstra vs. Bellman-Ford)
- Recognize real-world challenges: **count-to-infinity, poison reverse, hot potato routing**

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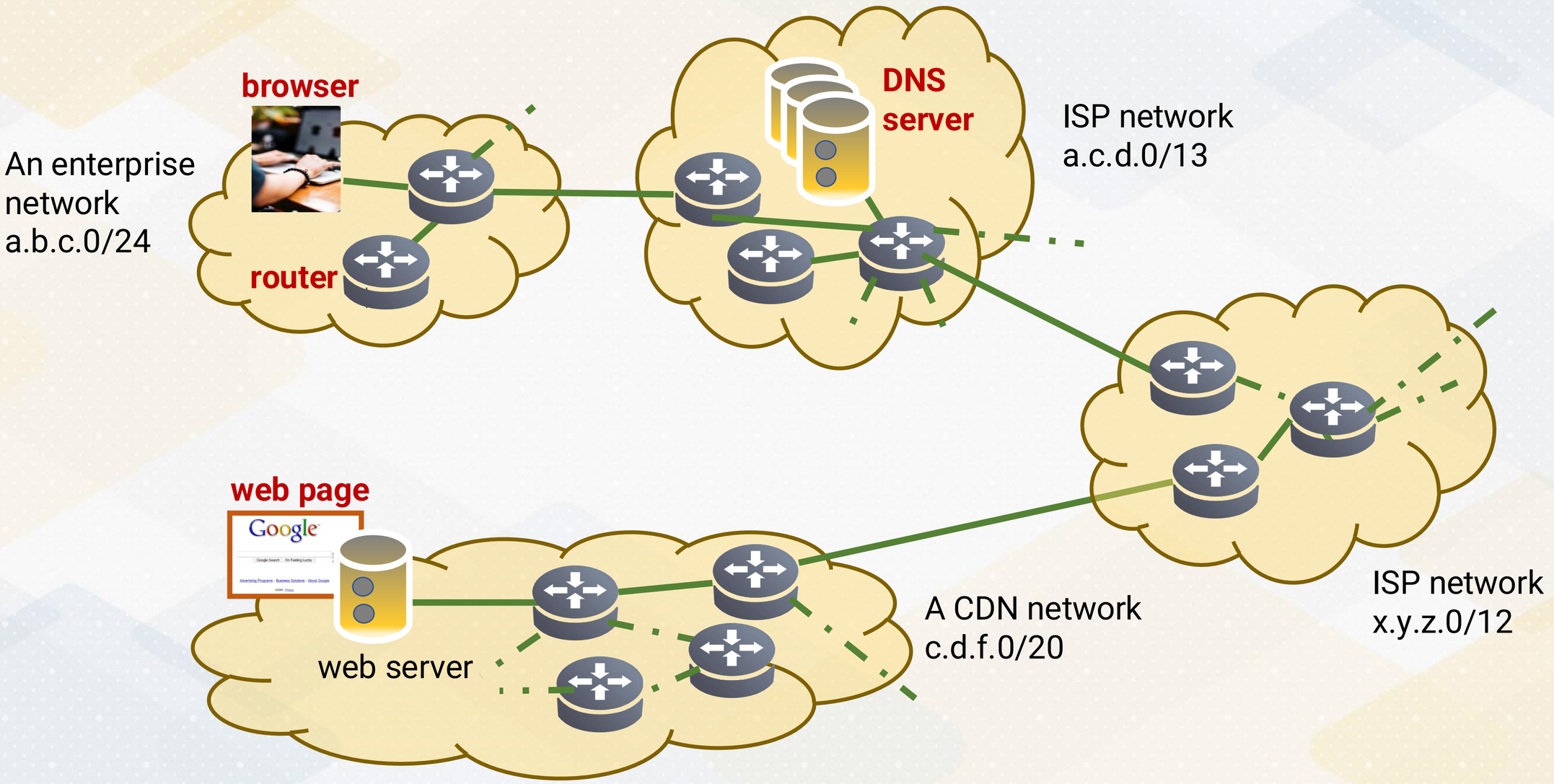
Reviewing the Internet's Protocol Stack



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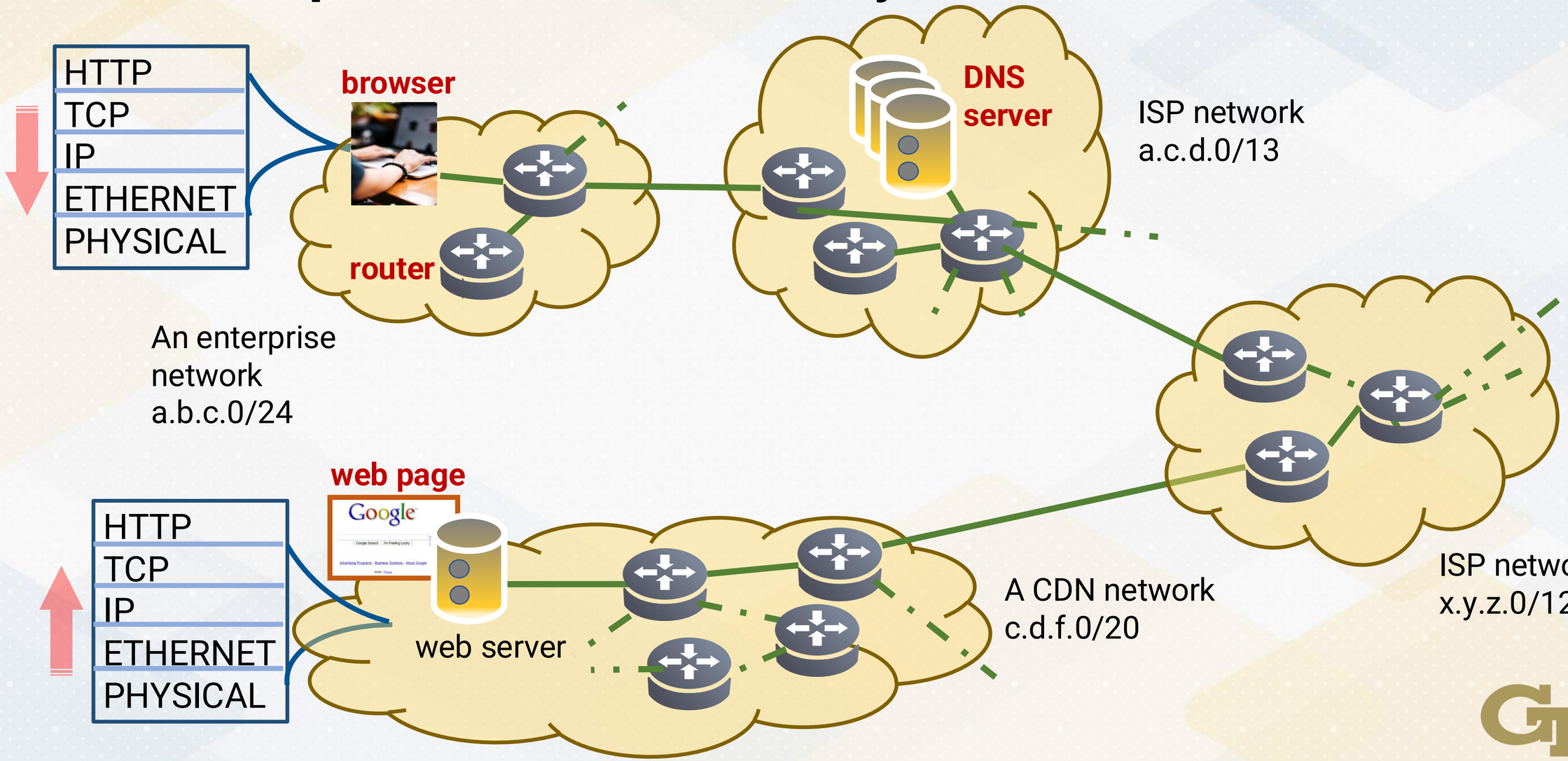
Reviewing the Internet's Protocol Stack
From the Transport to the Network Layer



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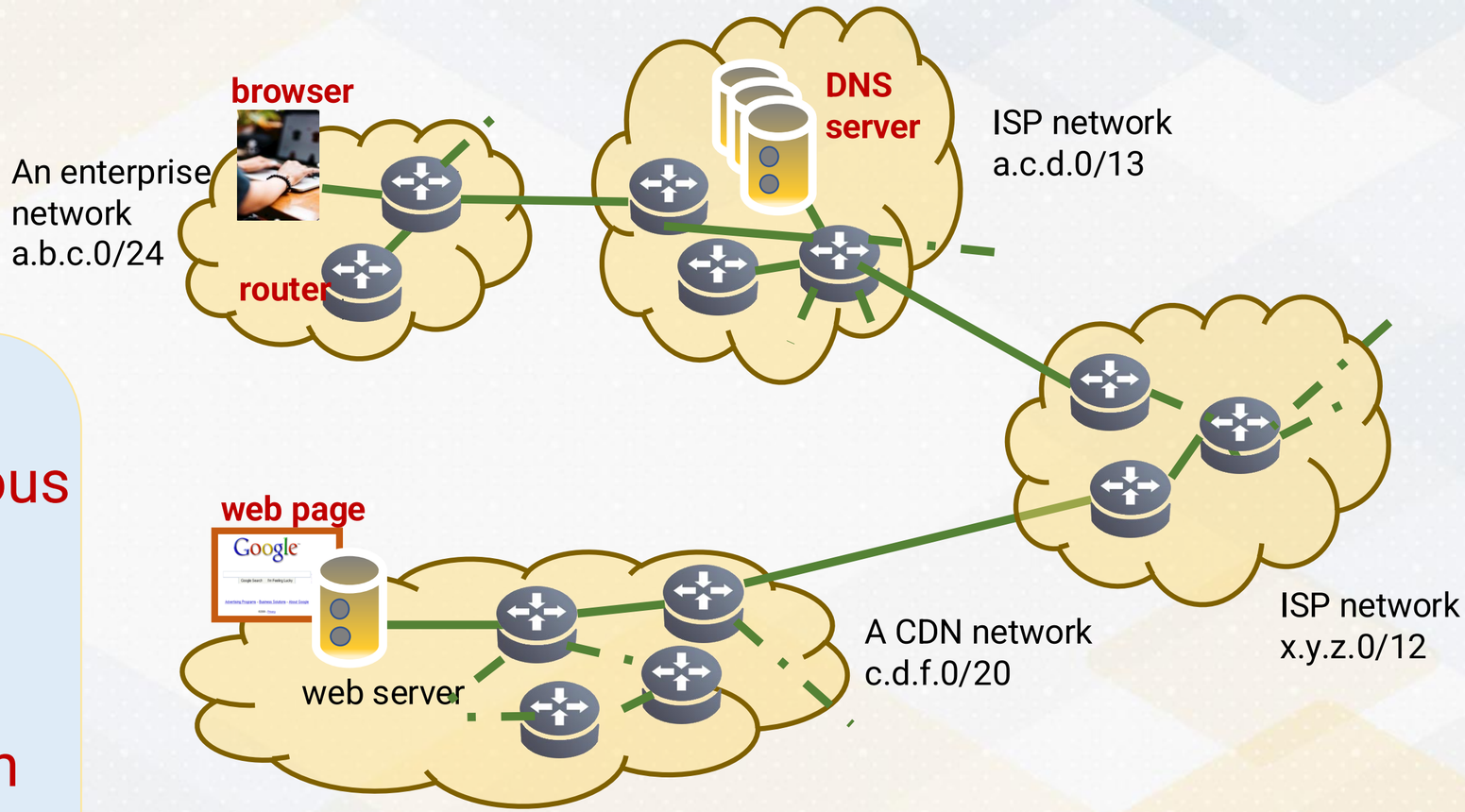
Reviewing the Internet's Protocol Stack
Traffic Travels Through Multiple ASes



- End-to-end traffic crosses multiple Autonomous Systems (ASes)
- Each AS makes routing decisions **independently**
- Two routing tasks: within an AS vs between ASes

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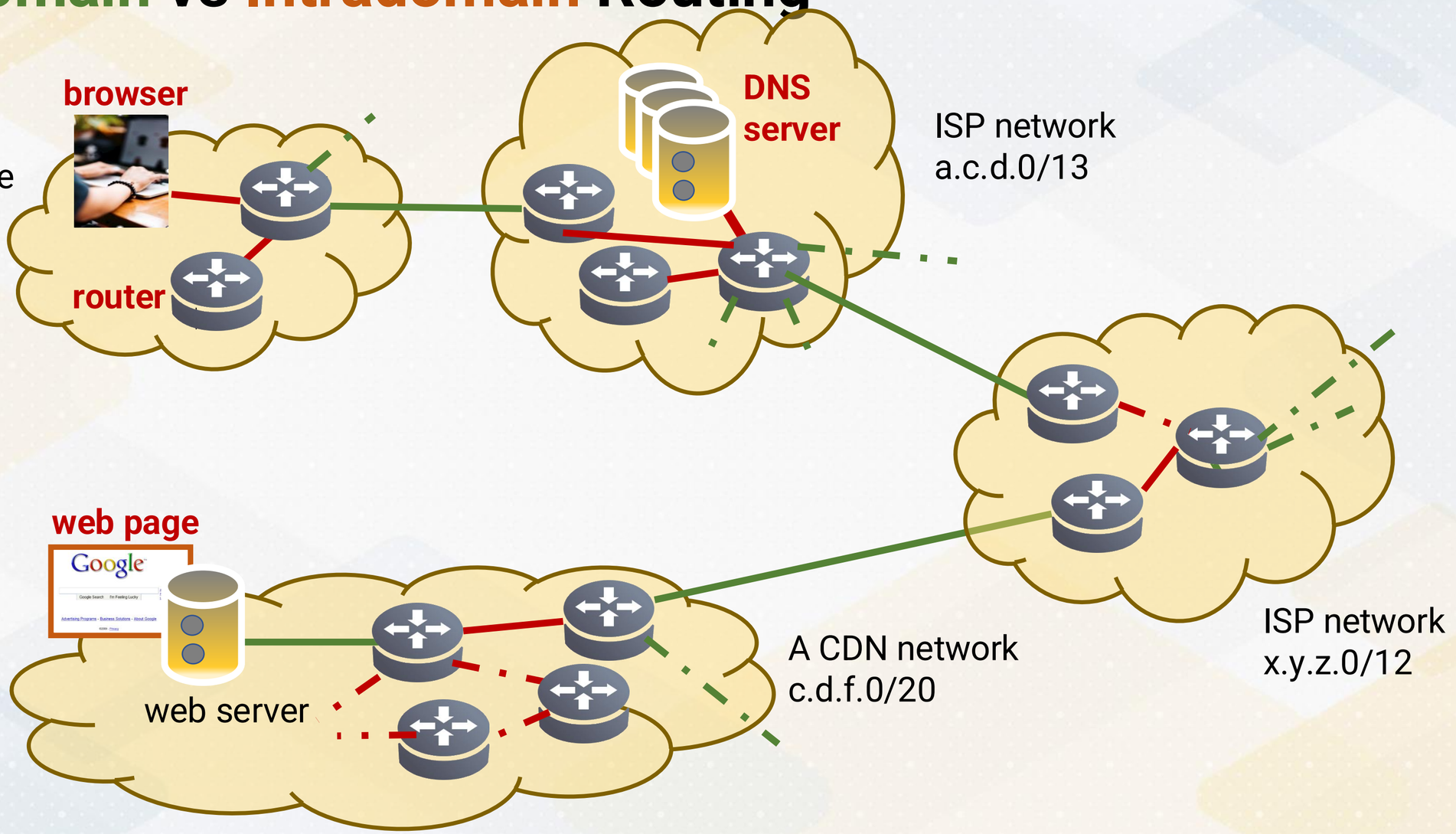
The Network Layer

Interdomain vs Intradomain Routing



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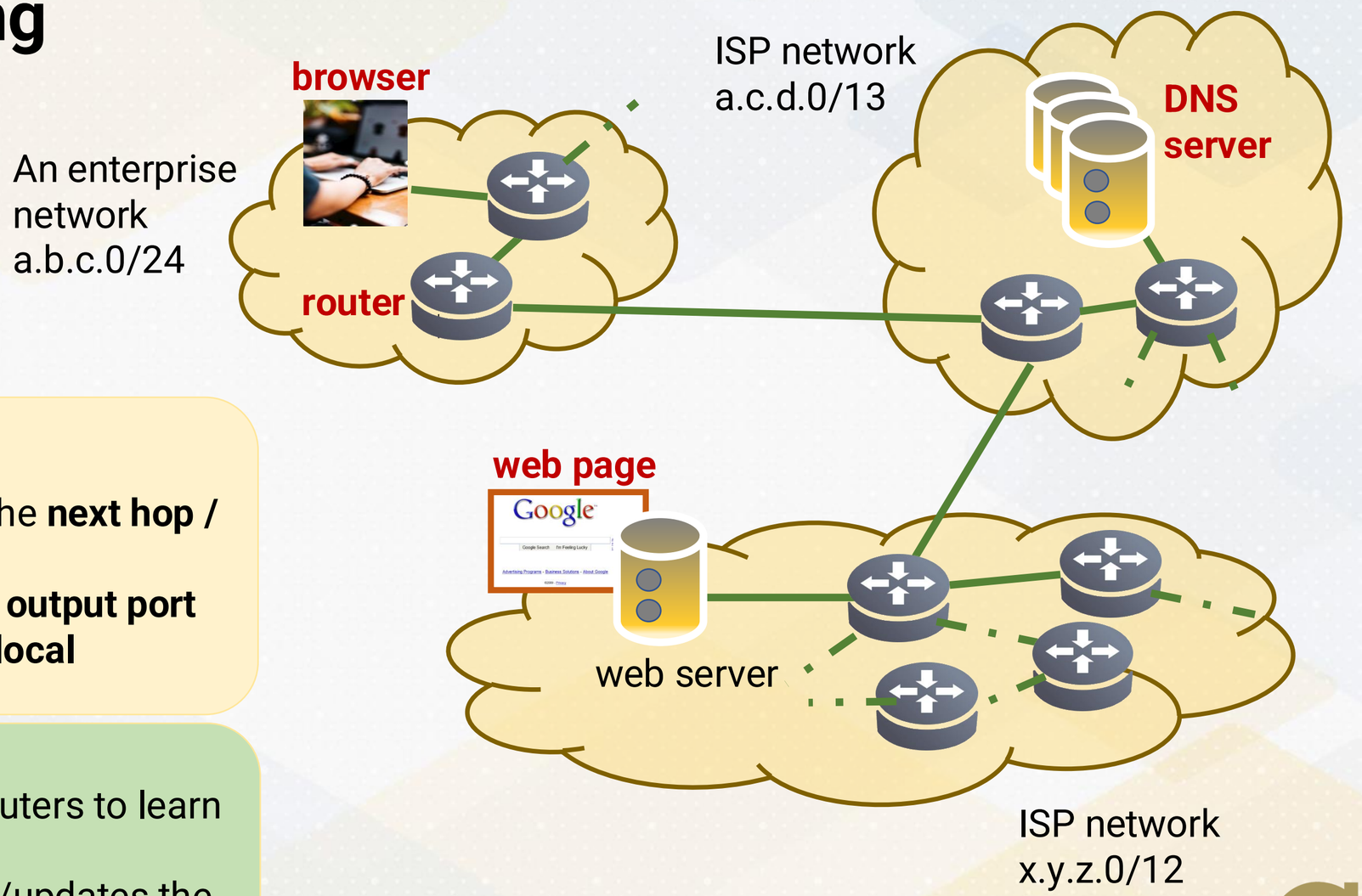
The Router's Tasks
Routing & Forwarding



Forwarding (data plane)

- Uses the forwarding table to select the **next hop / outgoing interface**
- Moves the packet from **input port to output port**
- Operations are **per packet, fast and local**

Routing (control plane)

- Exchanges information with other routers to learn about **paths/topology**
- Computes the **best paths** and builds/updates the forwarding table
- Adapts to **changes** (failures, cost updates)

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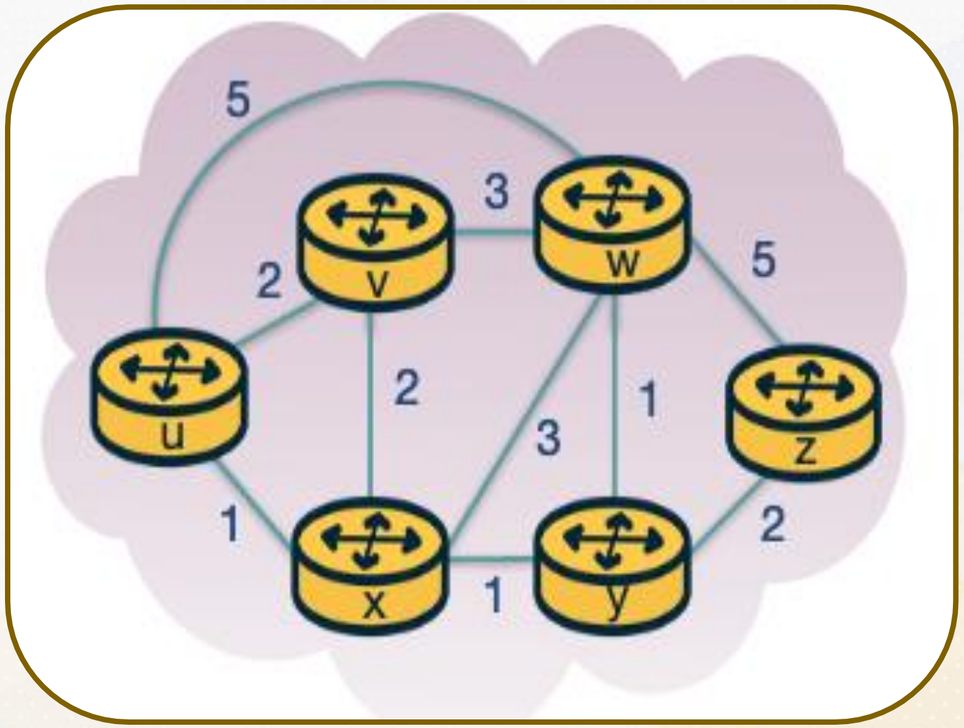
Link Stating Routing (1/3)

- Routers share a **common view of the network topology** (who is connected to whom)
- Each link has a **cost metric** (delay, bandwidth, admin-defined cost)
- A **path** is the sequence of routers/links a packet follows from source to destination
- The **best path** is the path with the lowest total cost (based on that metric)
- The **forwarding table** stores the **next hop** for each destination—the first step on the best path



Goal: use that shared map to find best paths and fill the forwarding table.

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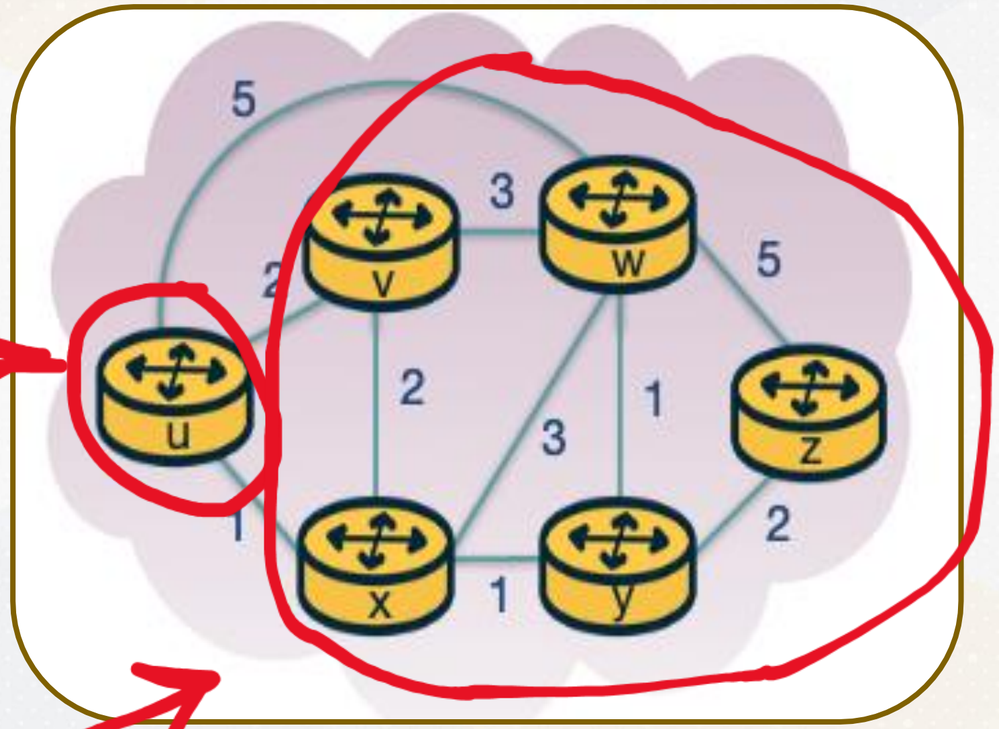
Link Stating Routing (2/3)

U: source node

V: every other node in the network



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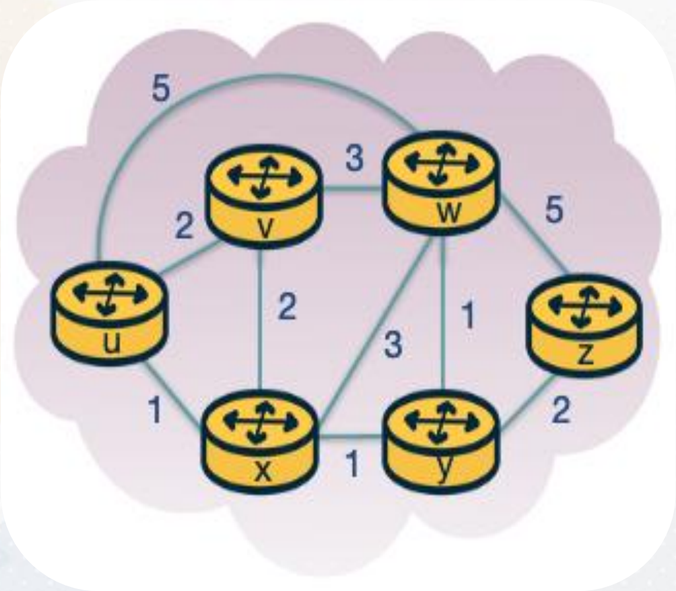
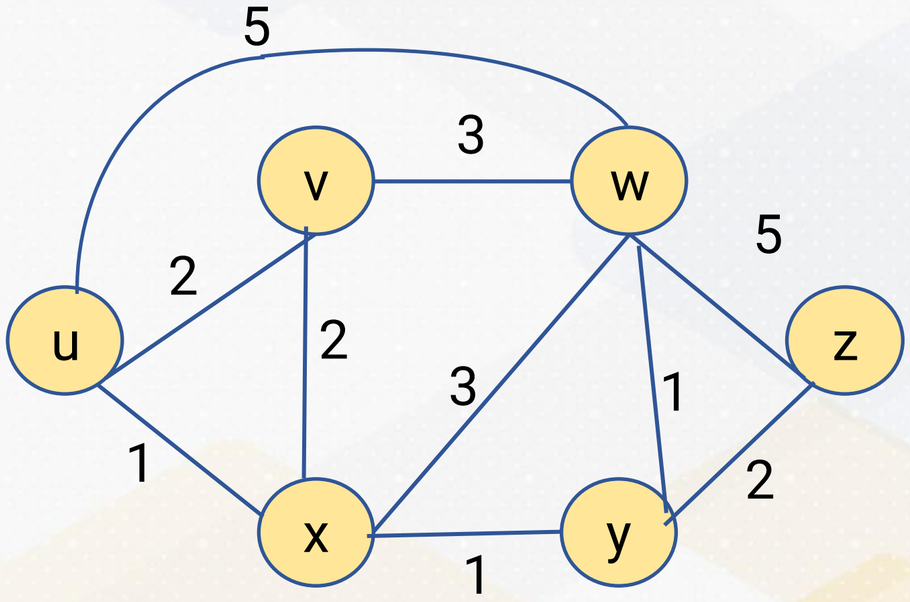
Link Stating Routing (3/3)



Graph representation



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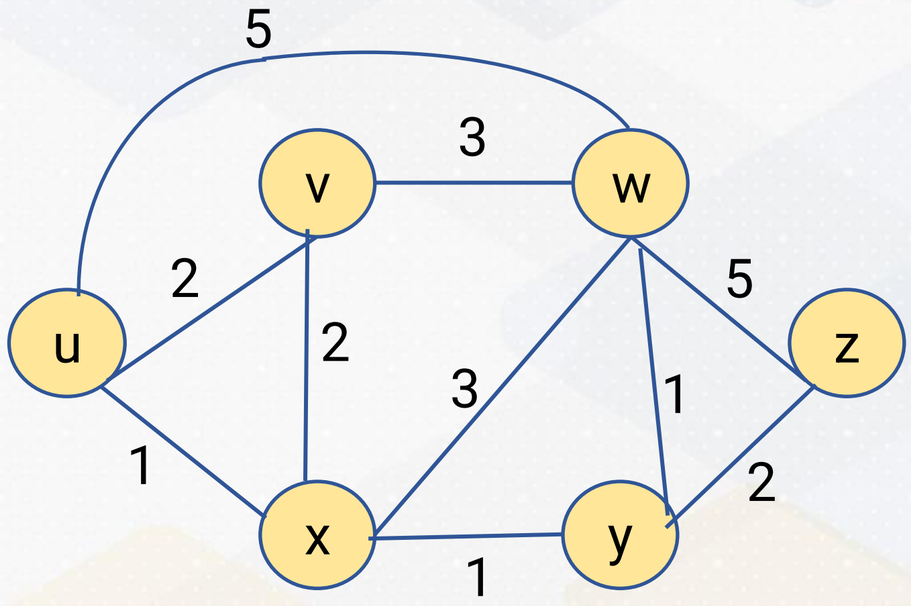
Link Stating Routing (1/3)
Dijkstra's Algorithm

- Keep track of **D(v)**: current best-known cost from source **u** to each node **v**
- Keep track of **p(v)**: predecessor on the best-known path (to reconstruct the route)
- Keep track of **N'**: set of nodes whose shortest paths from **u** are confirmed
- **Initialize:** N' = {u}; set D(neighbor) = link cost, and D(all others) = ∞ (with p(neighbor)=u)



step	N’	D(v),p(v)	D(w),p(w)	D(x),p(x)	D(y),p(y)	D(z),p(z)

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Link Stating Routing (2/3)
Dijkstra's Algorithm

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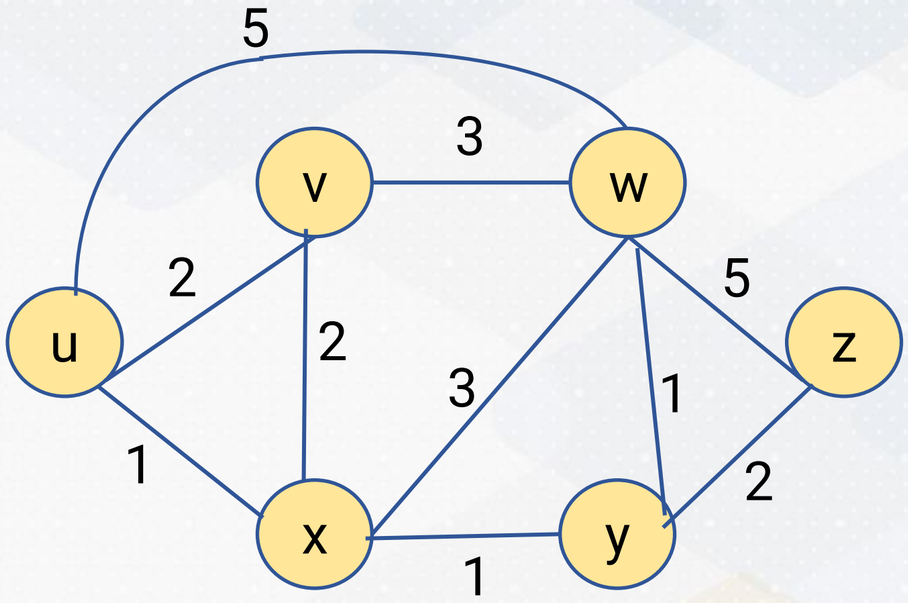

step	N’	D(v),p(v)	D(w),p(w)	D(x),p(x)	D(y),p(y)	D(z),p(z)
0	u	2, u	5, u	1, u	∞	∞

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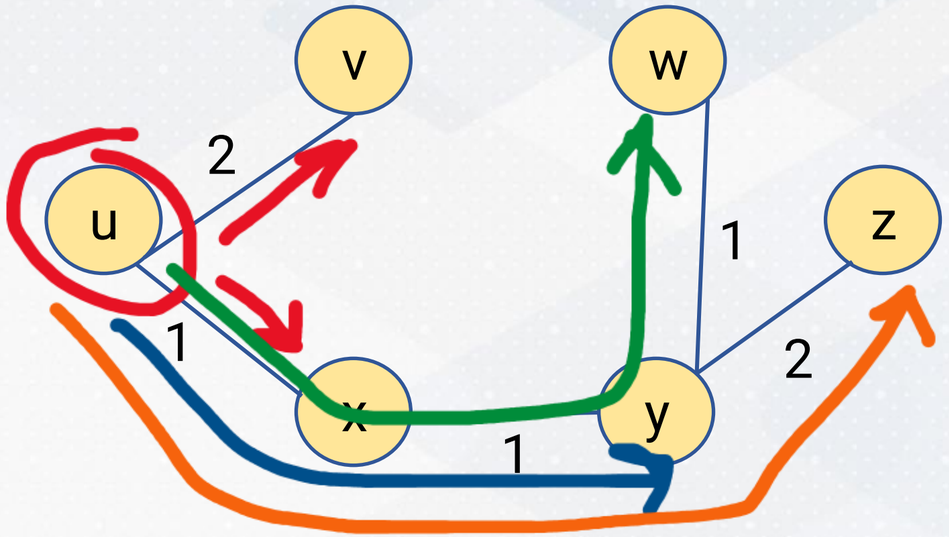
Link Stating Routing (3/3)
Dijkstra's Algorithm

- **When it ends**: we're done once **all nodes are in N'**, meaning the shortest path from the source to every destination is confirmed
- **Final result**: the algorithm produces a **shortest-path tree** rooted at the source; the router then builds the **forwarding table** by taking, for each destination, the **first hop** on that path
- **Complexity (why O(N²)):** Scan N−1, N−2, …, 1 nodes to find the next minimum ⇒ total ≈ N(N−1)/2.



step	N'	D(v),p(v)	D(w),p(w)	D(x),p(x)	D(y),p(y)	D(z),p(z)
0	u	2, u	5, u	1, u	∞	∞
1	ux	2, u	4, x		2,x	∞
2	uxy	2, u	3, y			4, y
3	uxyv		3,y			4,y
4	uxyvw					4,y
5	uxyvwz					

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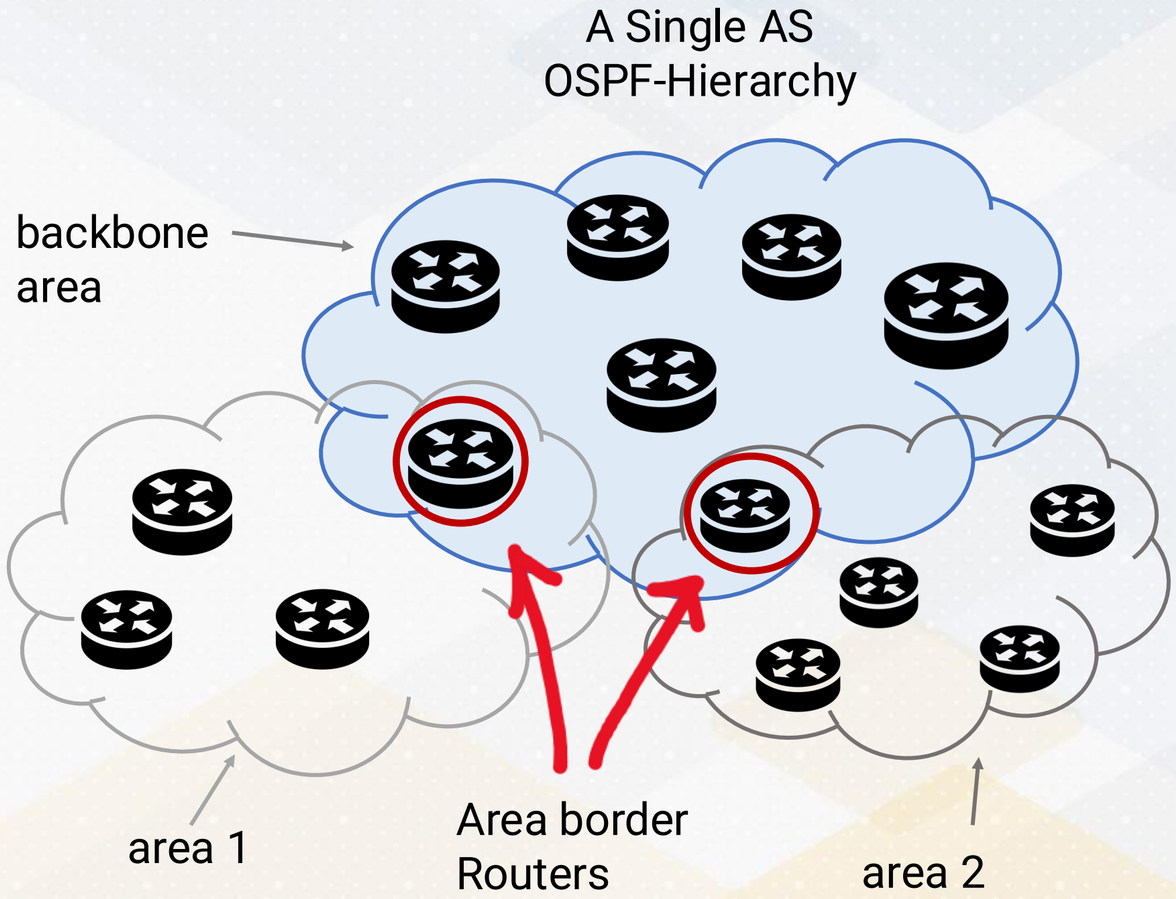
Link Stating Routing
OSPF

- OSPF is **an intradomain** (IGP, internal gateway routing protocol**) link-state protocol** used inside a single administrative domain (an AS)
- **Hierarchy for scale**: OSPF uses areas + a backbone, connected by Area Border Routers (ABRs)
- Routers flood **link-state advertisements (LSAs)** so routers in an area build the same topology database
- **Each router runs Dijkstra locally** and installs next hops into its forwarding table



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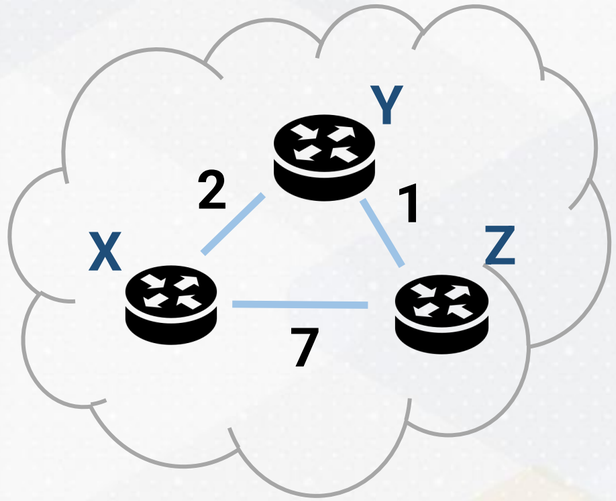
Distance Vector Routing (1/3)

- **Goal**: compute least-cost routes **without** a global network map
- Each router keeps **a distance vector**: its current cost to every destination
- **Routers exchange vectors with neighbors only**; information "diffuses" hop-by-hop



Bellman-Ford equation: $D_x(y) = \min_v\{c(x,v) + D_v(y)\}$

X's best cost to destination y is the minimum, over all neighbors v, of (cost to reach v) + (v's best cost to reach y)

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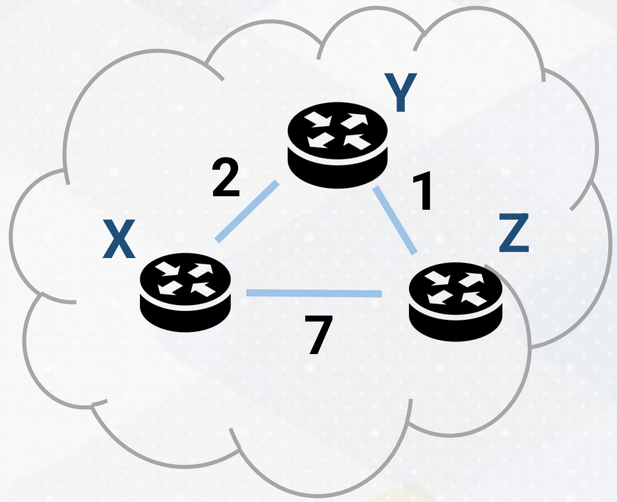
Distance Vector Routing (2/3)

- Routers exchange **distance vectors with neighbors**
- Update rule: **cost via neighbor = link cost + neighbor's advertised cost**
- If a best **cost changes**, **update the table and re-advertise**
- Information spreads **hop-by-hop through the network**



Node x table

From \ Cost to	x	y	z
x	0	2	7
y	∞	∞	∞
z	∞	∞	∞

Node y table

From \ Cost to	x	y	z
x	∞	∞	∞
y	2	0	1
z	∞	∞	∞

Node z table

From \ Cost to	x	y	z
x	∞	∞	∞
y	∞	∞	∞
z	7	1	0

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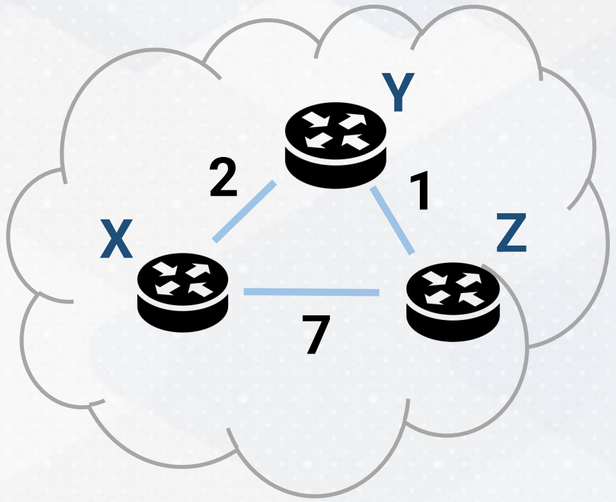
Distance Vector Routing (3/3)

- **Converges** when no router has any new updates to send
- **Final state**: each router knows its **least-cost distance** to every destination
- Practical output: choose the **next hop** that achieves that best cost
- Example: x→z becomes **3 via y** (better than the direct cost 7)



Node x table

From \ Cost to	x	y	z
x	0	2	3
y	2	0	1
z	3	1	0

Node y table

From \ Cost to	x	y	z
x	0	2	3
y	2	0	1
z	3	1	0

Node z table

From \ Cost to	x	y	z
x	0	2	3
y	2	0	1
z	3	1	0

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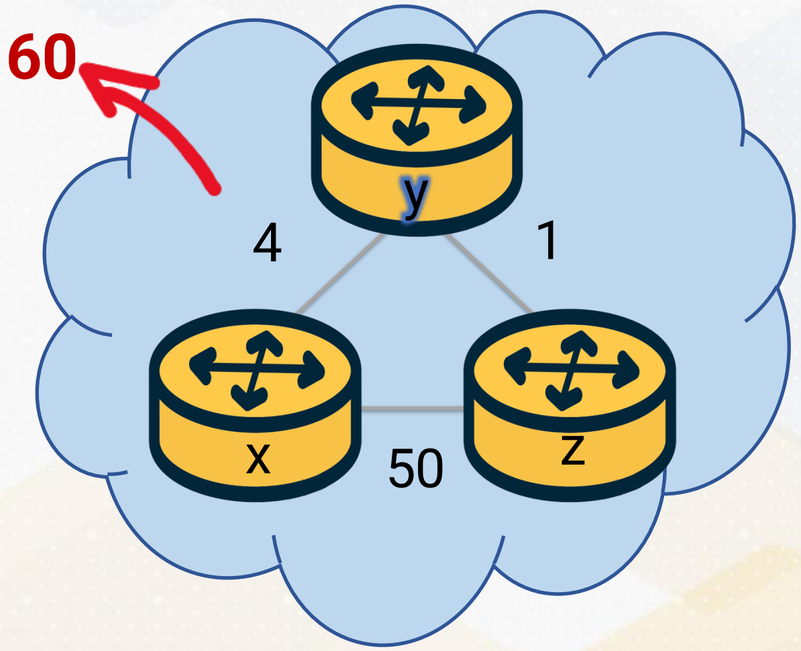
Distance Vector Routing
Count to Infinity

- **Trigger: a link cost increases** (or a link fails), “**bad news travels slowly**” in DV.
- Neighbors can form a temporary loop by trusting each other’s outdated distances

- After failure: Y routes to X via Z; Z routes to X via Y (mutual misconception)
- Routers **increment the advertised cost step-by-step** (6, 7, 8, …) toward “infinity”

- **Convergence can take many rounds** until a router switches to a **real alternative** (or hits a max metric)



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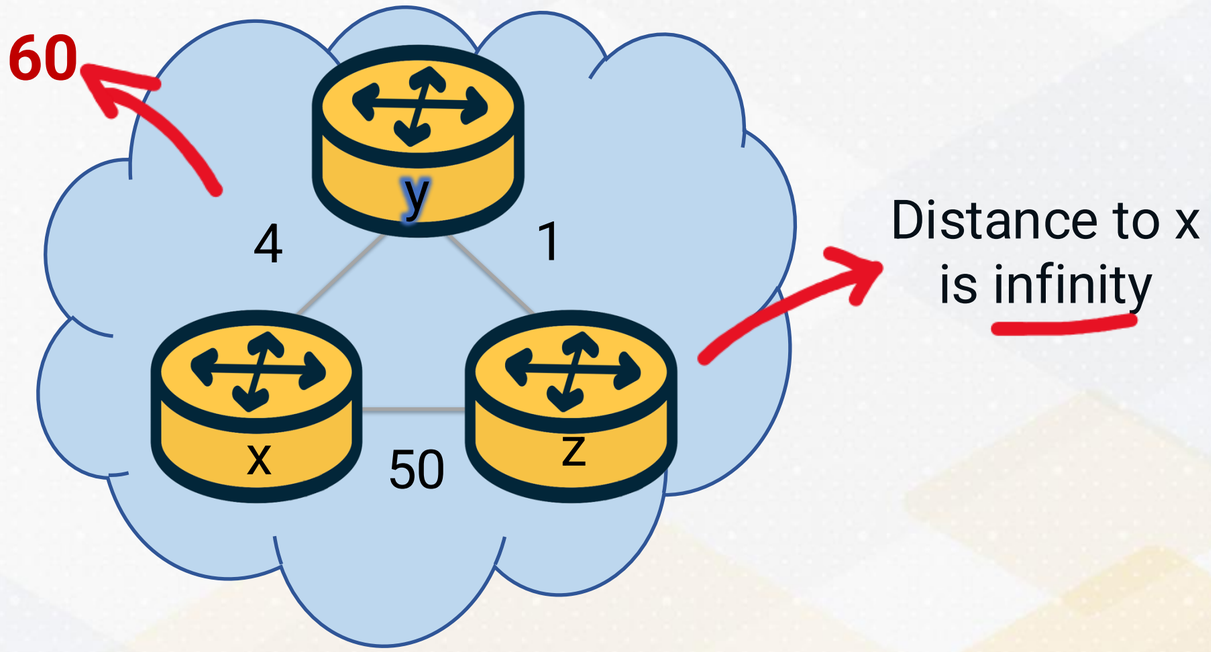
Distance Vector Routing
Count to Infinity → Poisson Reverse

- **If I route to destination x via neighbor y, I tell y: "my distance to x is ∞"**

- **Prevents two-node loops** (e.g. "you route through me, I route through you")
- **Speeds up convergence** after failures/cost increases in common cases
- **Limitation**: doesn't fully fix multi-node count-to-infinity scenarios



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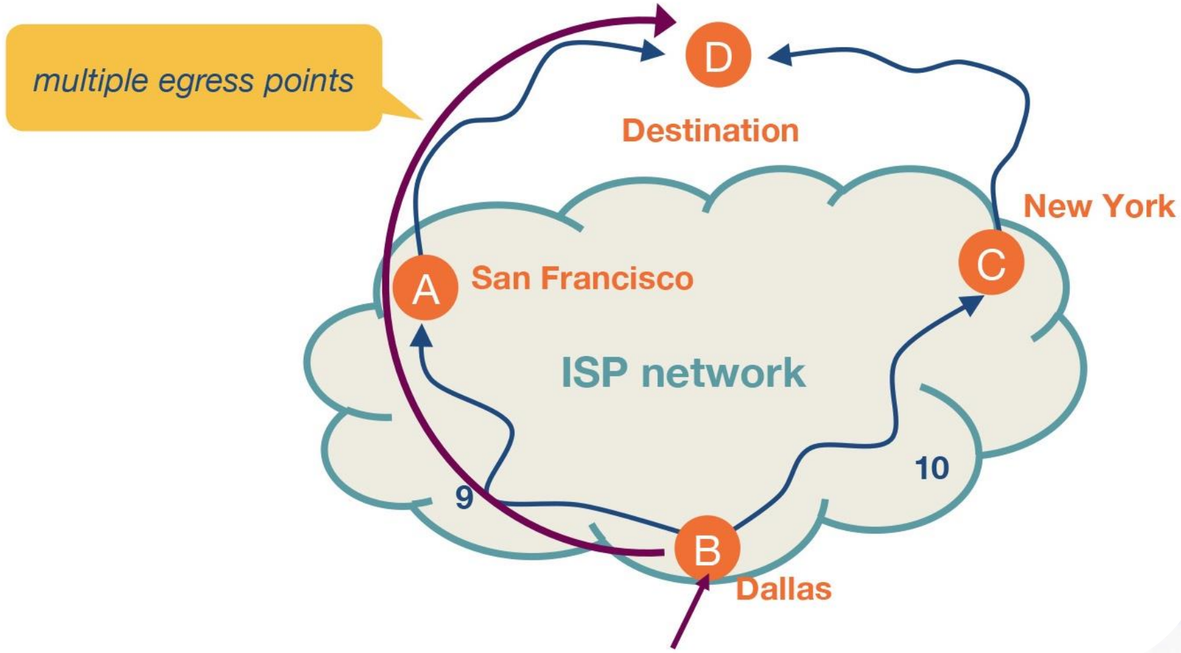
Hot Potato Routing

When Intra Can Affect Inter-domain Routing

From intra to inter: once a destination is *outside* the domain, the router must choose an **egress point.** That is an intradomain choice that directly shapes the interdomain.

Hot Potato Routing Rule: pick the closest exit based on **intradomain (IGP) cost** to reach that egress (“closest-exit routing”)
Motivation: hand traffic off quickly and reduce internal bandwidth or operational load (“get it out of my network”)
Impact: small IGP changes can shift the chosen exit and cause **traffic shifts.**



Teixeira, R., Shaikh, A., Griffin, T., & Rexford, J. (2004, June). Dynamics of hot-potato routing in IP networks. In *Proceedings of the joint international conference on Measurement and modeling of computer systems* (pp. 307-319).

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Summary

- TCP/UDP manage end-to-end delivery; IP carries packets router-to-router
- Inside a router: forwarding = per-packet lookup and send; routing = compute/maintain the lookup table
- Across the Internet: packets traverse multiple organizations (enterprise to ISPs to CDNs), so routing spans within and across administrative domains
- Intradomain routing approaches:
  - Link-state: flood link info, run Dijkstra, install next hops
  - Distance-vector: trade vectors with neighbors, apply Bellman-Ford, converge over iterations
- In practice: convergence issues (count-to-infinity, poison reverse), and policy-driven behavior -> closest-exit (hot potato) routing

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